LINE EQUATIONS

ANALYTIC GEOMETRY

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

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ENDTERM

TERM GRADE

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EXAM

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Equation of a Line

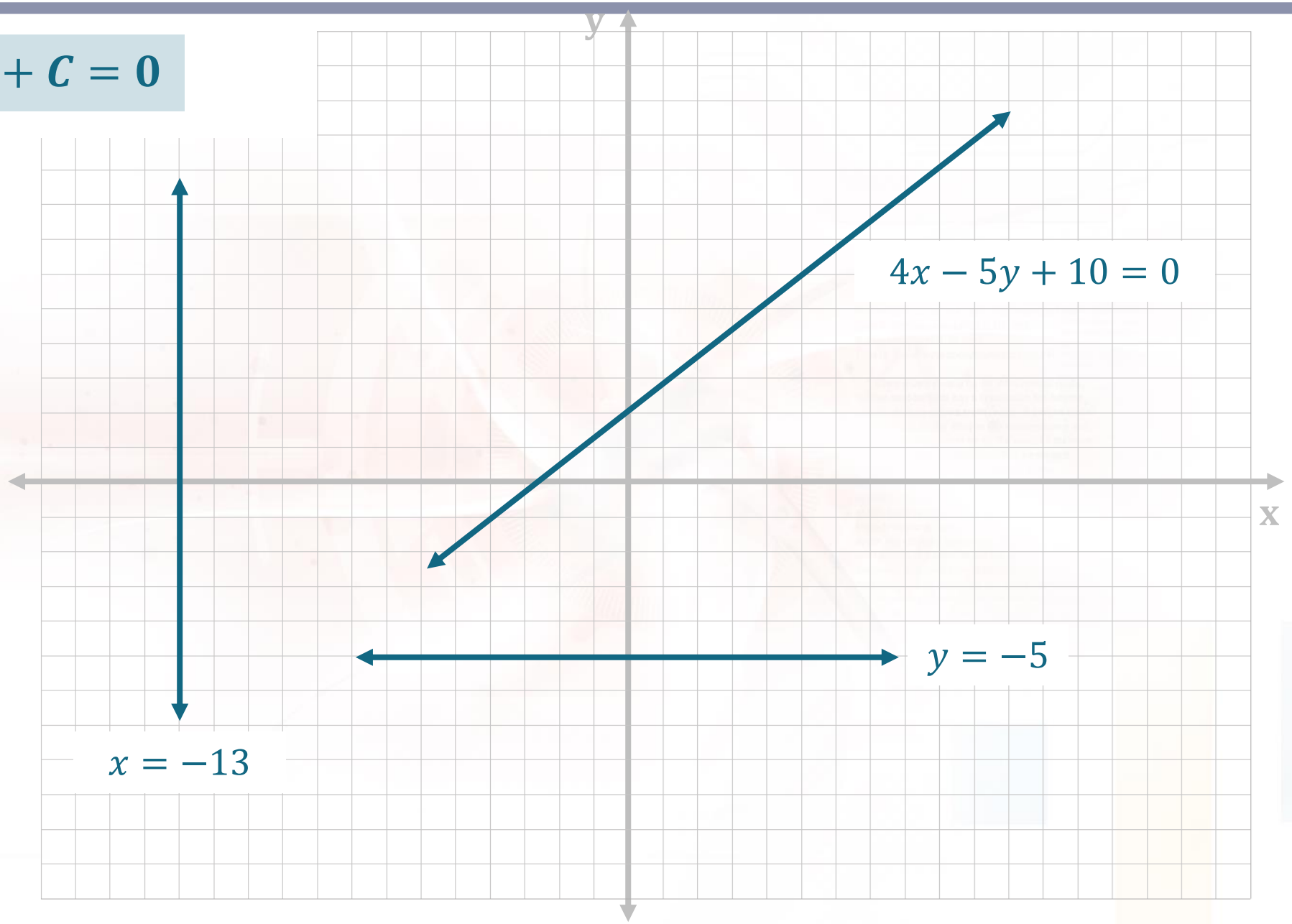
- General Equation
- Point Slope Form
- Slope Intercept Form
- Intercept Form

Parallel and Perpendicular Lines

EQUATIONS OF A LINE

GENERAL EQUATION

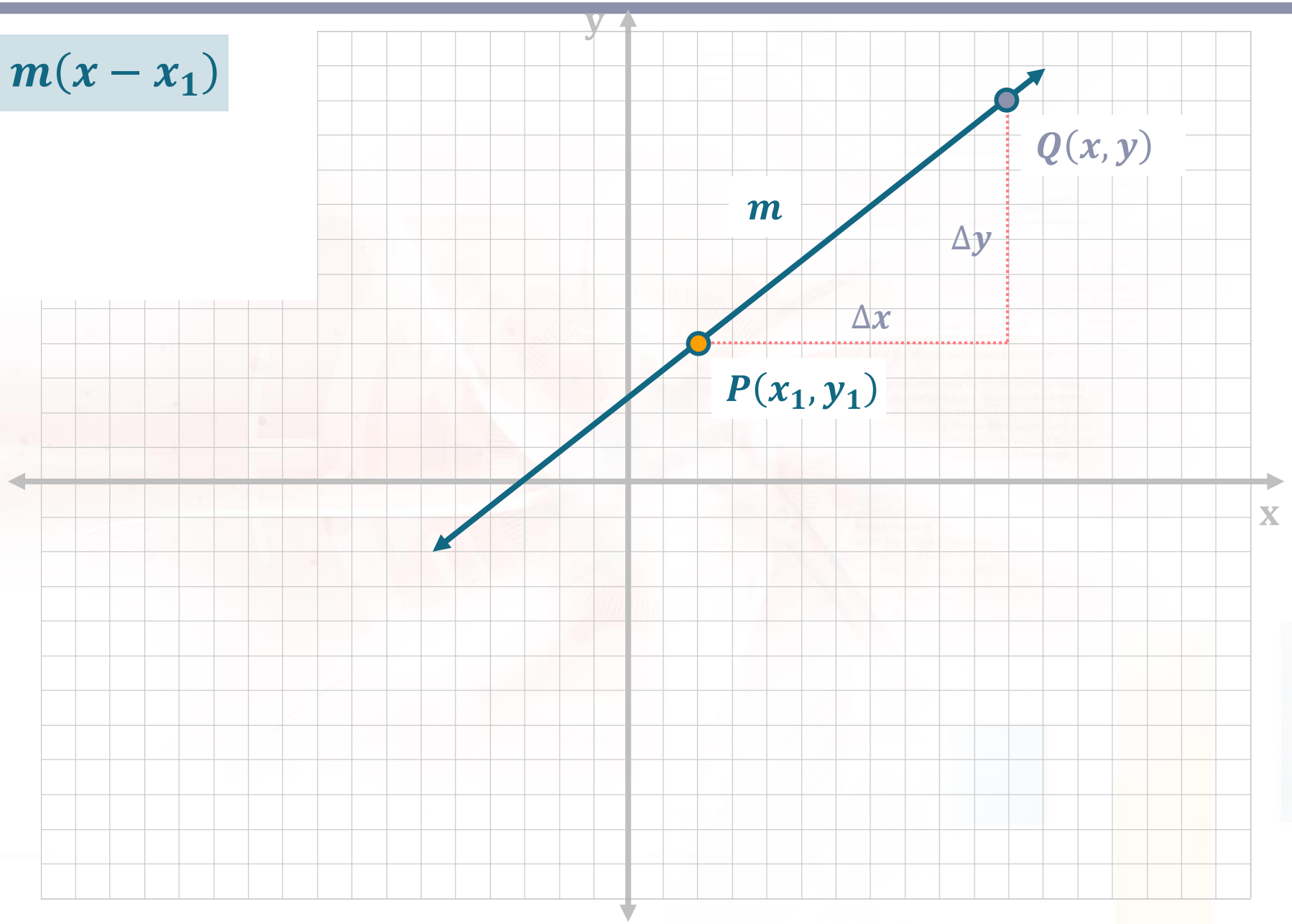
$$Ax + By + C = 0$$



POINT SLOPE FORM

$$y - y_1 = m(x - x_1)$$

$$m = \frac{\Delta y}{\Delta x}$$



Find the **equation** of the line through

$(2, -7)$ with **slope** $\frac{2}{5}$

$$m = \frac{2}{5}$$

$$y - y_1 = m(x - x_1)$$

$$y - (-7) = \frac{2}{5}(x - 2)$$

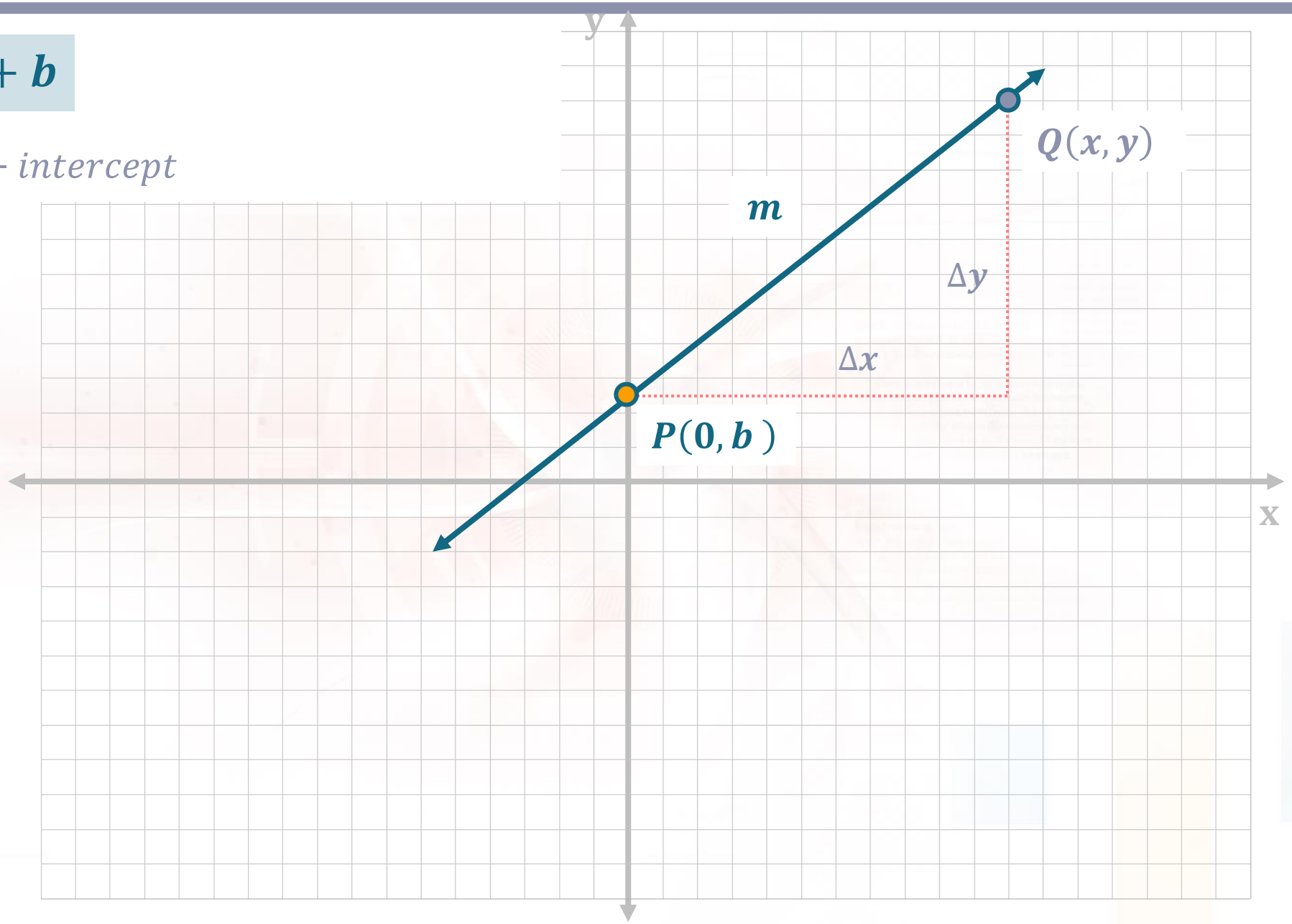
$$\boxed{y + 7 = \frac{2}{5}(x - 2)}$$

ans

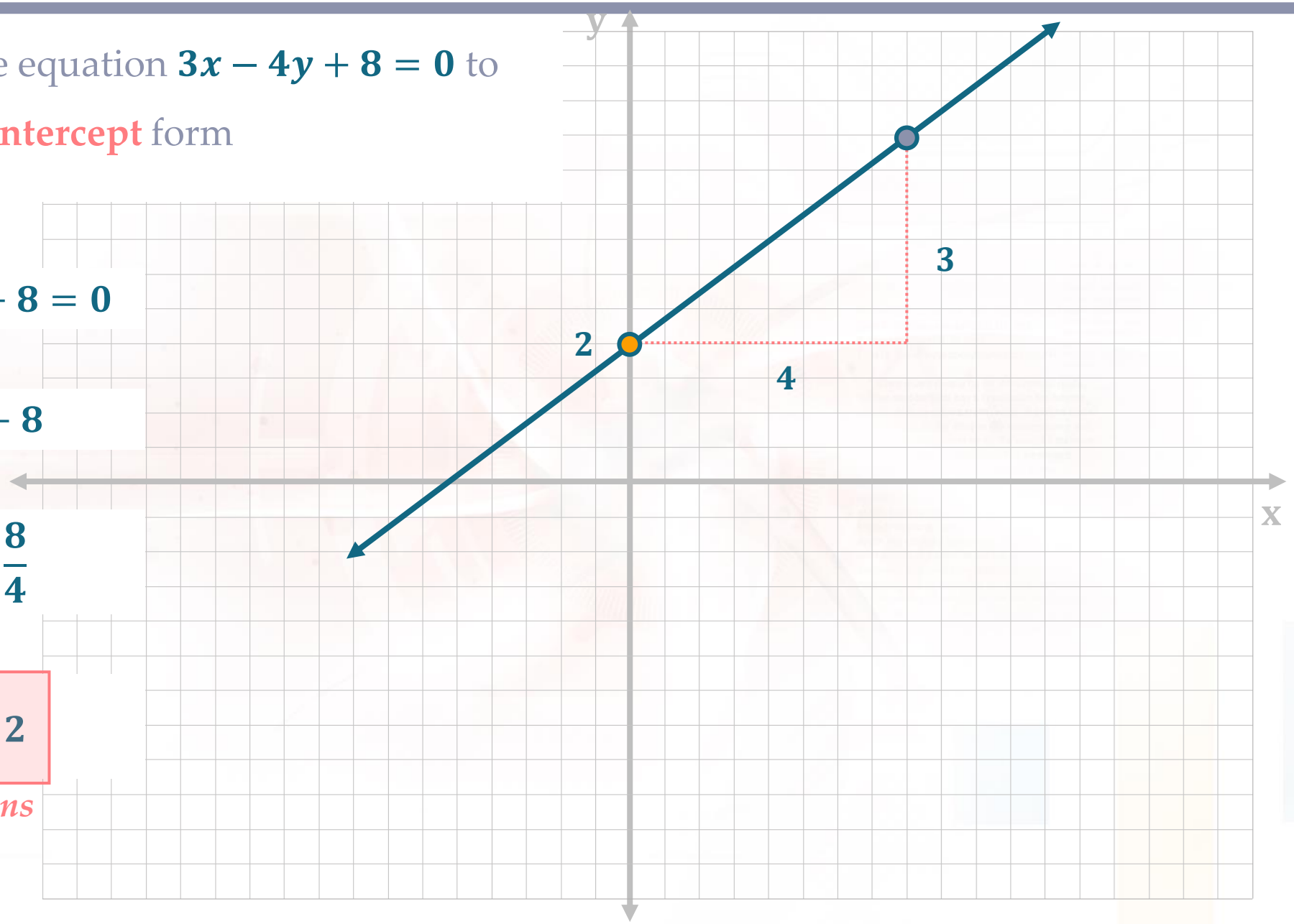
SLOPE INTERCEPT FORM

$$y = mx + b$$

b is the $y - intercept$



Reduce the equation $3x - 4y + 8 = 0$ to the **slope intercept** form



$$3x - 4y + 8 = 0$$

$$4y = 3x + 8$$

$$y = \frac{3}{4}x + \frac{8}{4}$$

$$\boxed{y = \frac{3}{4}x + 2}$$

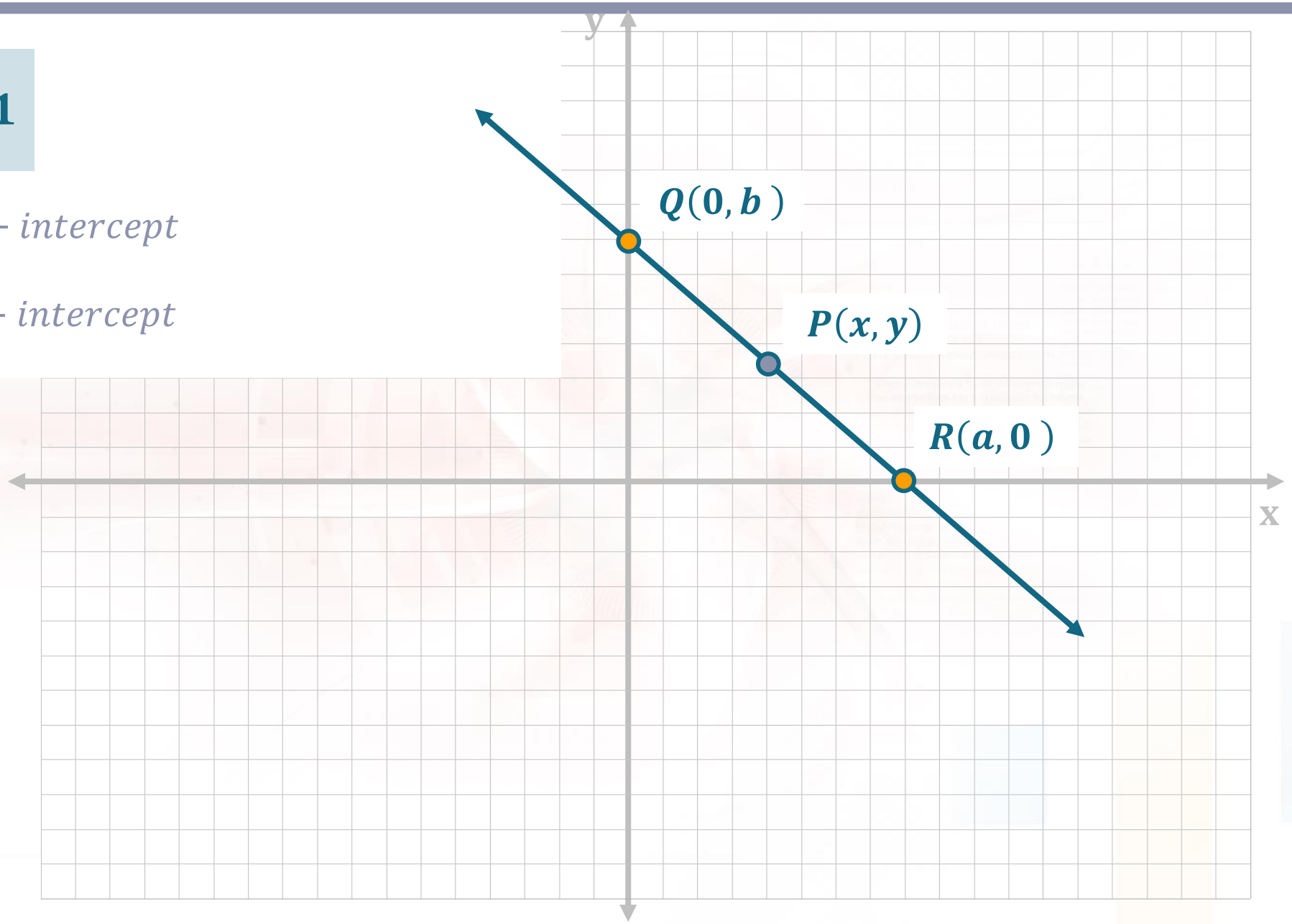
ans

INTERCEPT FORM

$$\frac{x}{a} + \frac{y}{b} = 1$$

a is the $x - intercept$

b is the $y - intercept$



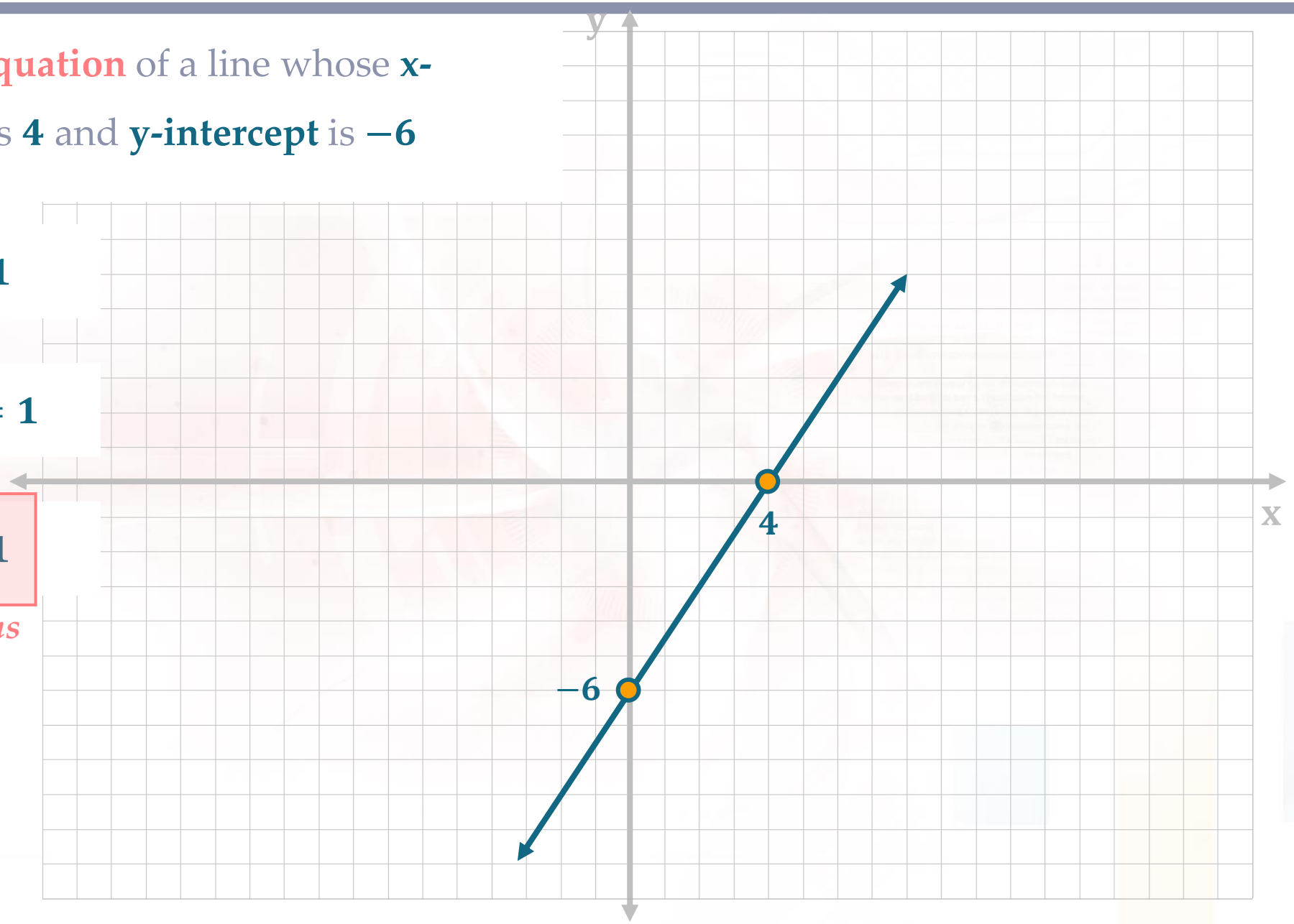
Find the **equation** of a line whose **x-intercept** is **4** and **y-intercept** is **−6**

$$\frac{x}{a}+\frac{y}{b}=1$$

$$\frac{x}{4}+\frac{y}{-6}=1$$

$$\boxed{\frac{x}{4}-\frac{y}{6}=1}$$

ans

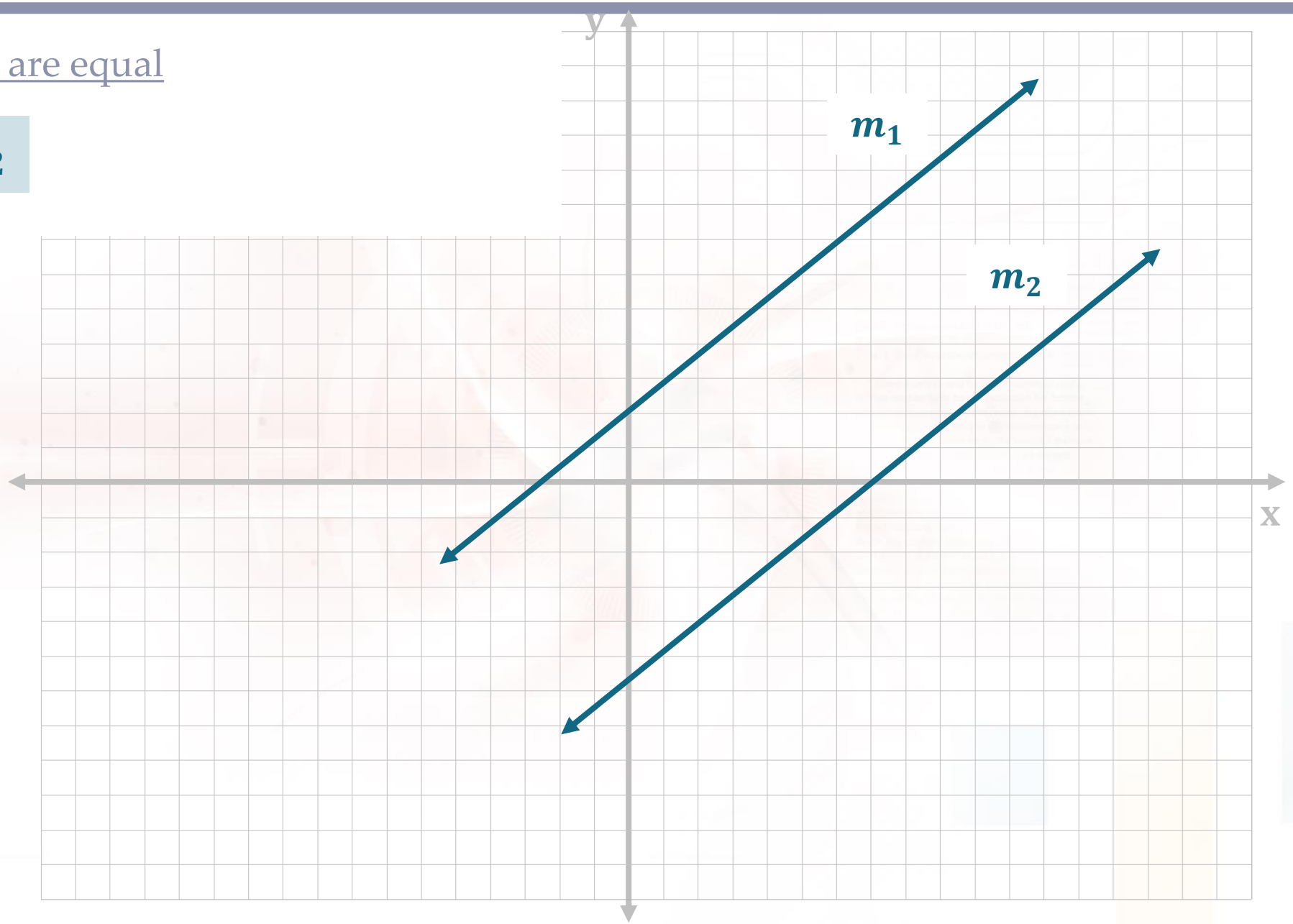


PARALLEL AND PERPENDICULAR LINES

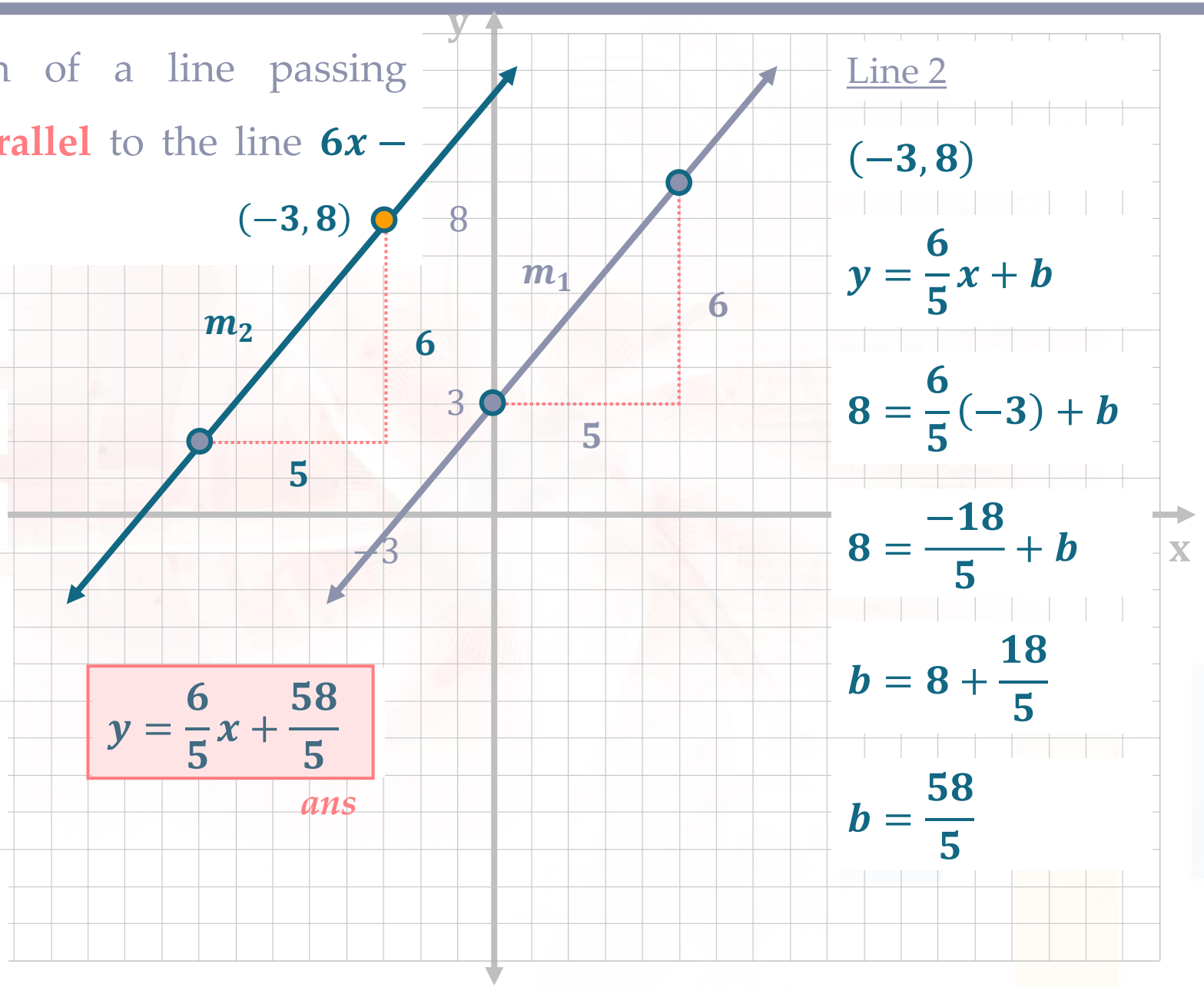
PARALLEL LINES

The slopes are equal

$$m_1 = m_2$$



Write the equation of a line passing through $(-3, 8)$, **parallel** to the line $6x - 5y + 15 = 0$



Line 1

$$6x - 5y + 15 = 0$$

$$5y = 6x + 15$$

$$y = \frac{6}{5}x + \frac{15}{5}$$

$$y = \frac{6}{5}x + 3$$

$$m_1 = m_2 = \frac{6}{5}$$

Line 2

$$(-3, 8)$$

$$y = \frac{6}{5}x + b$$

$$8 = \frac{6}{5}(-3) + b$$

$$8 = \frac{-18}{5} + b$$

$$b = 8 + \frac{18}{5}$$

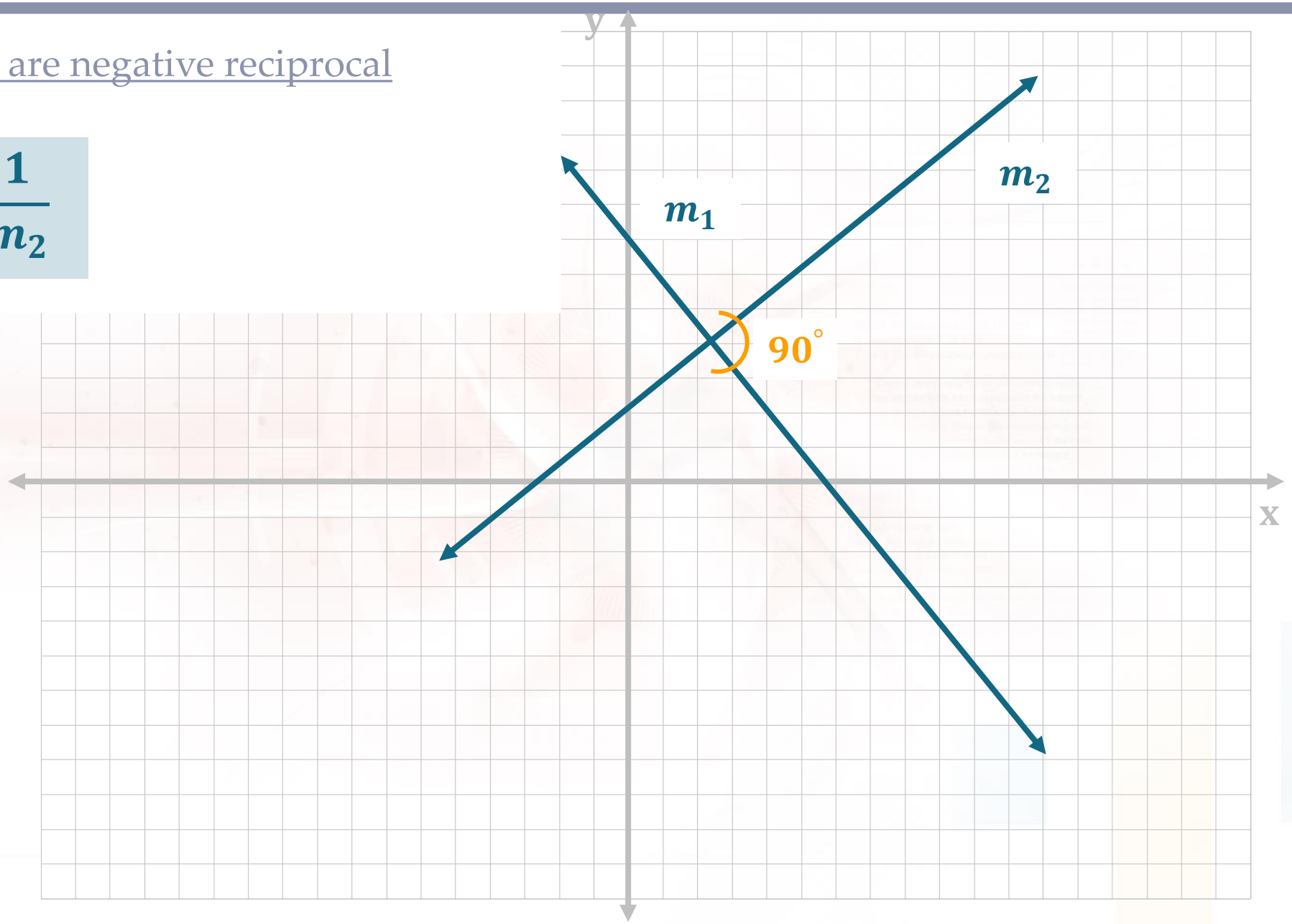
$$b = \frac{58}{5}$$

$$\boxed{y = \frac{6}{5}x + \frac{58}{5}}$$

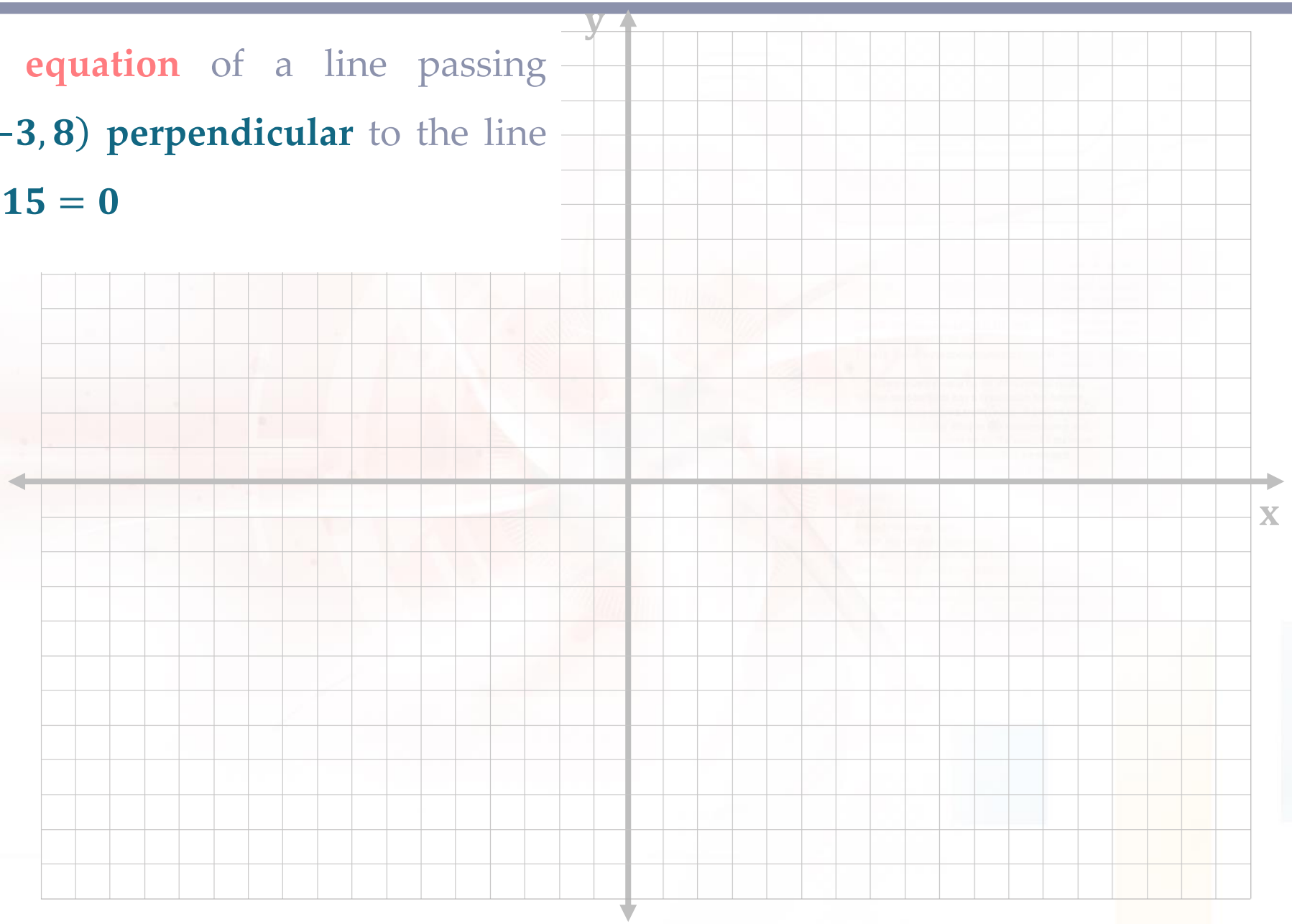
ans

The slopes are negative reciprocal

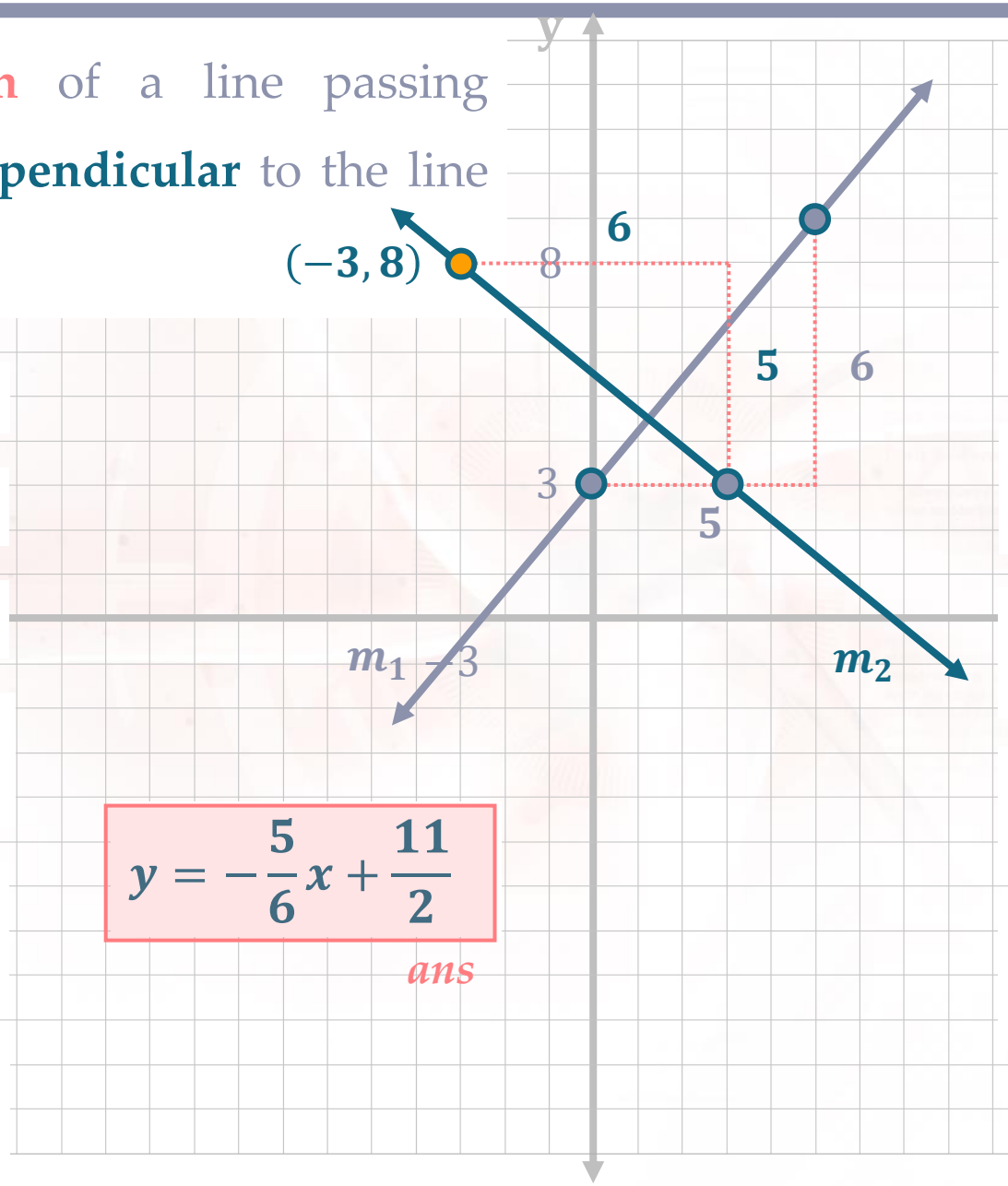
$$m_1 = -\frac{1}{m_2}$$



Write the **equation** of a line passing through $(-3, 8)$ **perpendicular** to the line $6x - 5y + 15 = 0$



Write the **equation** of a line passing through $(-3, 8)$ **perpendicular** to the line $6x - 5y + 15 = 0$



Line 1

$$6x - 5y + 15 = 0$$

$$5y = 6x + 15$$

$$y = \frac{6}{5}x + \frac{15}{5}$$

$$y = \frac{6}{5}x + 3$$

$$m_1 = \frac{6}{5}$$

Line 2

$$(-3, 8)$$

$$m_2 = -\frac{5}{6}$$

$$y = -\frac{5}{6}x + b$$

$$8 = -\frac{5}{6}(-3) + b$$

$$8 = \frac{15}{6} + b$$

$$b = 8 - \frac{15}{6}$$

$$b = \frac{11}{2}$$

$$\boxed{y = -\frac{5}{6}x + \frac{11}{2}}$$

ans

SEATWORK